

CSE5026: Introduction and Frontiers of Cognitive Science

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Cognitive Science 认知科学

- Understanding how *human mind* works is important to a broad range of human activities.
- 理解心智如何工作对实践活动至关重要，在AI时代尤为重要
- Teachers need to understand how students' mind
- Engineers need to understand users' mind
- Decision makers need to understand people's mind
- AI/Agents need to understand users' mind

Learning Objectives

- Develop a comprehensive understanding of the theoretical foundations, research questions, and methodologies of cognitive science.
- Develop a vision and interest in relation to modern artificial intelligence.
- 全面了解认知科学的理论基础、主要研究内容、研究方法。
- 培养对人工智能相关的前沿问题的兴趣与视野。

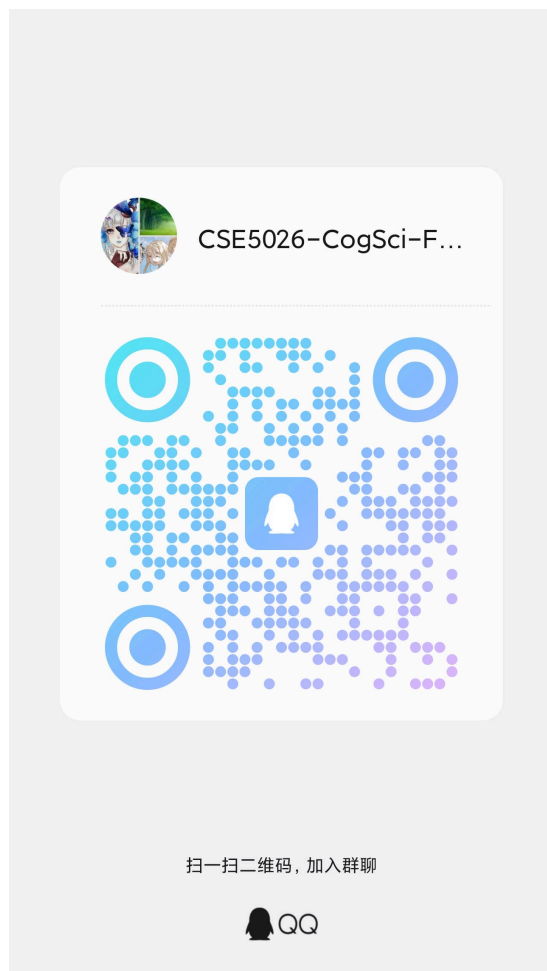
Course Information

- Class meeting: Tuesday, 10:20 am, 3-4
- Lab session: Wednesday, 10:20 am, 3-4
- Optional Textbook: 《认知科学 Cognitive Science》 史忠植, 中国科学技术大学出版社。
- Optional online materials:
 - 中科院计算所, 智能科学与人工智能实验室, 认知基础。
<http://www.intsci.ac.cn/znjx/rzjc/>

Course website

- <https://clcs-sustech.github.io/zh/course/cse5026/>
- Slides and lab materials will be published and updated here.

课程群 QQ Group



- All course announcements will be made in this group.
- Graded lab practices and group project will be submitted to Blackboard.

Prerequisite

- Knowledge and concepts
 - CS303/CS311 Artificial Intelligence
- Paper reading and presentation skills
- Programming skills
 - Python, Numpy, PyTorch
- Math and statistics
 - Regression models and statistical tests

Grading Policies

- 10%: Attendance (5% class + 5% lab)
- 30%: Graded lab practice
 - All lab practices will be graded within a week's time.
 - 5 out of 15 will be placed higher weight (taking up to 25%)
- 20%: Paper reading and presentation
- 20%: Project
- 20%: Final exam

Late Policy

- 20% percent off per calendar day

Paper Reading and Presentation

- **One presentation report** for each team.
- Pick one topic from the current pool.
- Everyone (including other teams) needs to read the materials before the class.

Group Project

- Topics:
- EEG data analysis for human face perception.
- Eye-movement patterns during reading activities.
- Other cognitive science-related topics in LLM studies.

Team

- Team size: 2-3.
- Team up before the end of **week 3**.
- Paper assignment by the class meeting of **week 4**.
- Project topic decided by **week 8**.

Schedule Example

- Week 1: General introduction
 - The birth and development of cognitive science.
 - Research contents (what).
 - Research methods and technologies (how).
 - Relationship between CogSci and AI.
 - Programming tools: Python and libraries.
 - Statistical libraries.

Schedule Example

- Week 2: Neural Basics of Cognition
 - Neuron: Basic composition, synaptic transmission, neurotransmitters.
 - 神经元：基本组成，突触传递，神经递质
 - Resting membrane potential and action potential.
 - 静息膜电位与动作电位
 - Nervous system and cerebral cortex.
 - 神经系统与大脑皮层
 - Neuromorphic computing. 类脑计算理论
 - EEG Equipment. 脑电波设备

What is CogSci?

- Historical Background
- Research Methodology (theoretical)
- Research Methods and Technology (practical)
 - Tools; Experimental designs; Math and statistics.

Historical Background

- “Cognition”: 拉丁语词源 “cognoscere” : to learn, to recognize, to get to know.
 - 英义: The mental action or process of acquiring knowledge and understanding through thought, experience, and the senses.
 - 直译: 通过思考、经验和感官来获取知识和理解的过程。
- 认知/認知: 认 (认识) + 知 (知识)
 - 从外界认取, 转换为内在知识
- Closely related to psychology
- Psychology studies the **process** and **traits** of human minds
- Processes: dynamic courses; Traits: static features

Historical Background: Psychology

- It is difficult to discover "universal laws" in psychology
- Human body and mind are **complex** and **adaptive** to environment
- Even within the same environment, there are individual differences
- Different theories/schools (学派) of psychology to study complex human behaviors:
 - Behaviorism 行为主义, Gestalt 格式塔
 - Information Processing Theory 信息加工

Historical Background: Behaviorism 行为主义

- Behaviorism formula: Stimulus - Response (S-R) model

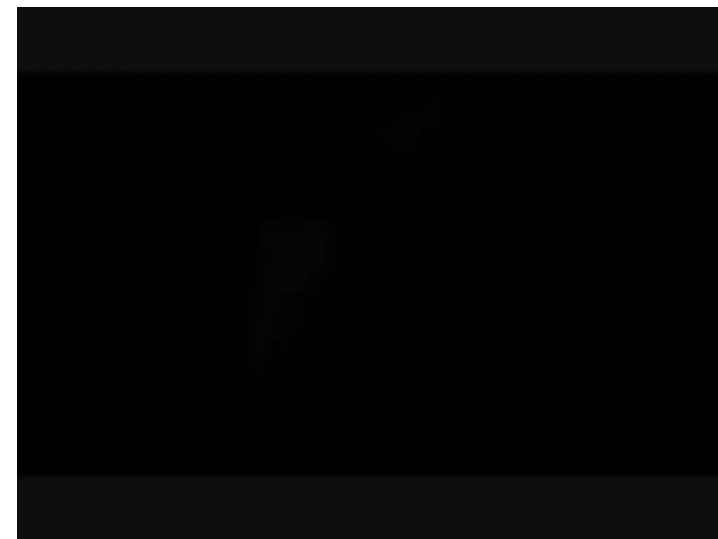
$$R = f(S)$$

- Response is the function of stimulus
- Russian scientist Pavlov's study on dogs 巴甫洛夫
- American psychologist J.B. Watson 华生, 1913
 - Little Albert Experiment: <https://www.youtube.com/watch?v=9hBfnXACsOI>
- American psychologist B.F. Skinner 斯金纳

Little Albert Experiment

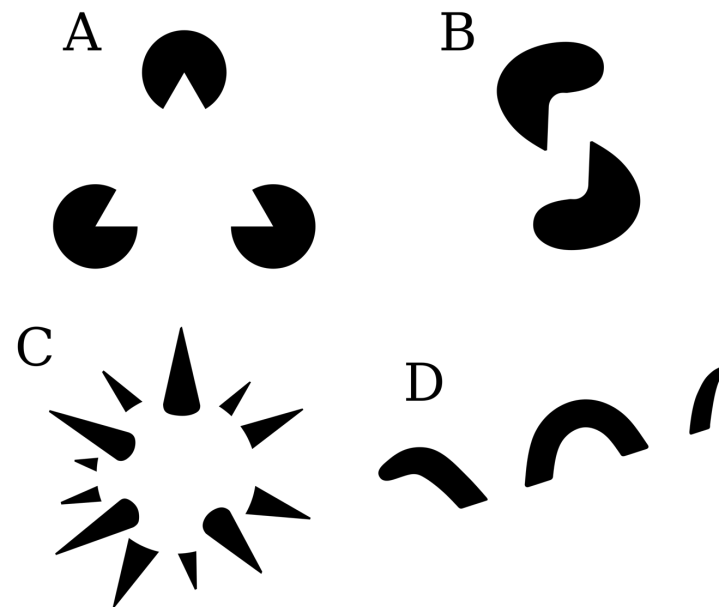
Source: <https://zh.wikipedia.org/zh-cn/小艾伯特实验>

- Albert: An 11 month old infant
- ① 巨响（非条件刺激）出现，引起恐惧（无条件反射）
Sound $\Rightarrow$ Fear (unconditioned reflex)
 - ② 白鼠（中性刺激）与巨响（非条件刺激）同时出现，引起恐惧（无条件反射）
Mouse + sound $\Rightarrow$ Fear (unconditioned reflex)
 - ③ 白鼠（条件刺激）出现，引起恐惧（条件反射）学习在此时发生。
Mouse $\Rightarrow$ Fear (conditioned)



Historical Background: Gestalt 格式塔

- Early 20th century, German psychologists Wertheimer, Koffka, Kohler.
- The **whole** of something is greater than the **sum** of its parts
- **Principles:**
 - Emergence 整体性/涌现性
 - Reification 具体化
 - Multistability 组织性
 - Invariance 恒常性



Source: <https://zh.wikipedia.org/wiki/格式塔学派>

Emergence of Cognitive Science

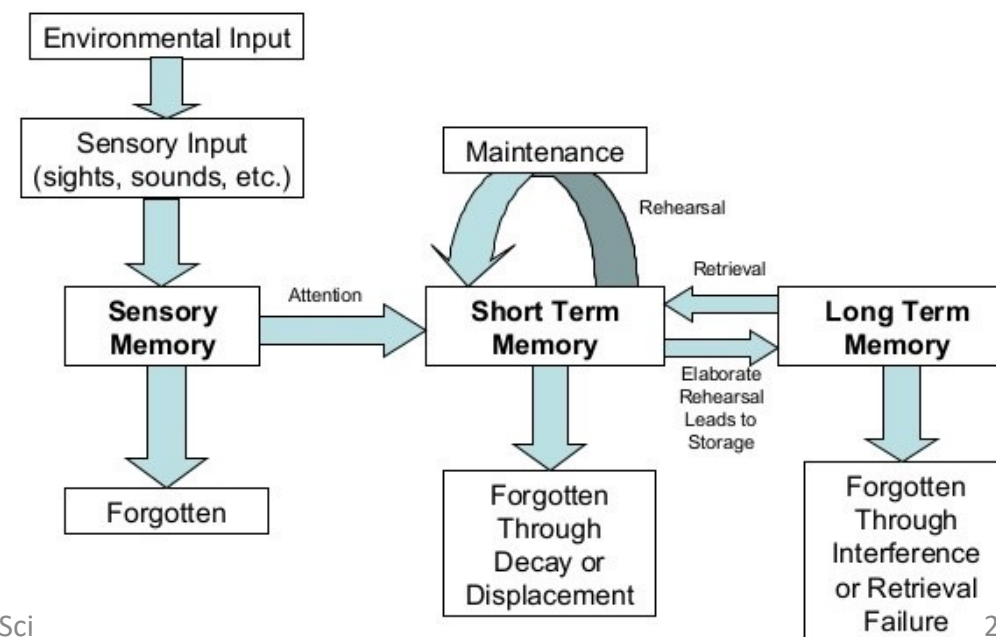
- The limitations of Behaviorism
 - J.B. Watson: 一切心理活动都归因为刺激与反应 S-R。
 - 心理学中唯一确实的、站得住脚的材料是机体的**行为**。
The only definitive and tenable material in psychology is the behavior of the organism.
(论迹不论心)
- Game changer: The booming of Information Technology
- **Information Processing Theory** 信息加工理论:
 - Human can be viewed as an information processing system
 - Perception, memory, and thought can be viewed as information production, reception and transmission.

Information Processing Theory

- Information 信息; Processing 加工
- Human mind is similar to a computer -- information is processed in **stages**, similar to how a computer *takes in* (input), *stores* (write), and *retrieves* (read) information.

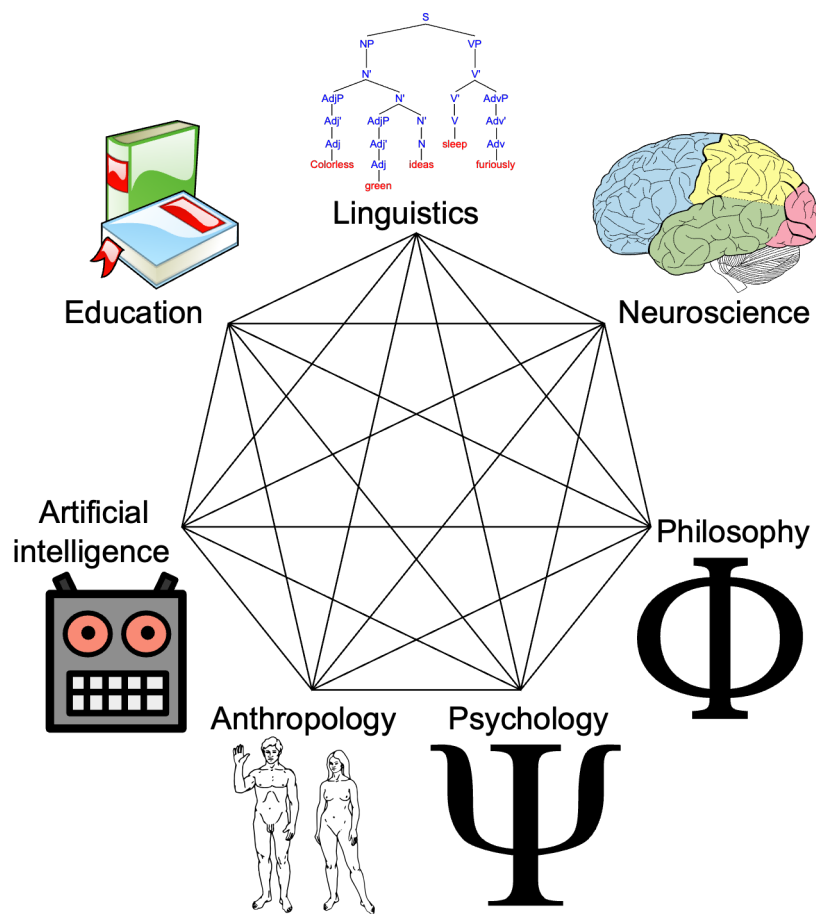
- Atkinson-Shiffrin memory model (1968) :

- Sensory memory 感知
- Short term memory 短期
- Long term memory 长期



CogSci as a Discipline

Source: https://en.wikipedia.org/wiki/Cognitive_science



Cognition is a science of mind and intelligence

- Perception
- Memory
- Thought & reasoning
- Learning
- Language
- Problem solving
- Emotion
- ... and more topics

What is CogSci?

- Historical Background
- **Research Methodology (theoretical)**
- Research Methods and Technology (practical)
 - Tools; Experimental designs; Math and statistics.

CogSci Research Methodology

- **Internalism 内在主义方法**
- 1. Reductionism 还原主义:
 - 高层性质和功能，可归结为低层的性质和功能
Higher-level properties and functions can be reduced to lower-level properties and functions.
 - 心理过程可还原为脑过程：心脑合一
Mental processes can be reduced to brain processes: mind and brain are one.
 - 例：脑损伤 ⇒ 影响人的思维、意识、气质、情感
Evidence: Brain injury ⇒ impacts human thought, consciousness, temperament, and emotion.
 - 困难：心理始终具有超物质性 (trans-material)，难以同生理、物理概念找到一一对应。

CogSci Research Methodology

- **Internalism 内在主义方法**
- 2. Functionalism 功能主义
- 强调心理活动的**功能**：心智是有机体与环境之间的中介。
Emphasizing the function of mental activity: the mind is the mediator between the organism and the environment.
- 智能的功能即解决问题，即有机体对环境的适应。
The function of intelligence is problem-solving—that is, the organism's adaptation to the environment.
- Hilary Putnam (1977) :
 - Mind (thinking) analogy to computer (computing)
 - Mental state must be explained using its functional relation with another state, just like computer behavior cannot be explained by the underlying physics/chemistry, but rather the higher level programs.
 - 计算机的出现使得我们对心智的认识可以摆脱还原论的限制
 - Source: Hilary Putnam on Functionalism https://youtu.be/i0bky2ep_ll

CogSci Research Methodology

- **Internalism 内在主义方法**
- 3. Introspectionism 内省主义
- Psychologists: Wilhelm Maximilian Wundt (威廉·冯特 1832-1920) and Edward Titchener (爱德华·铁钦纳 1867-1927)
- 通过系统性内省观察法解析意识经验的构成要素及其组合规律。
To analyze the constituent elements of conscious experience and the laws of their combination through the method of systematic introspection.

CogSci Research Methodology

- **Internalism 内在主义方法**

- 3. Introspectionism 内省主义

- (continued) 将心理现象分解为基本元素进行实验研究。

To decompose psychological phenomena into basic elements for experiment.

- 冯特首创实验内省法，要求被试在严格控制条件下报告自身意识经验的即时状态。

Wundt pioneered the method of experimental introspection, requiring subjects to report the immediate state of their conscious experience under strictly controlled conditions.

- 铁钦纳提出机械内省原则，要求研究者将复杂心理现象分解为感觉、意象、情感三种基本元素，并通过标准程序记录其强度。

Titchener proposed the principle of mechanical introspection, requiring researchers to decompose complex mental phenomena into three basic elements—sensations, images, and affections—and to record their intensity through standardized procedures.

- Now we also ask LLMs to self-report 😊

CogSci Research Methodology

- **Internalism 内在主义方法**
- 4. Cognitivism 认知主义 (心理学) / Symbolism 符号主义 (人工智能)
- 代表人物: Herber Simon, Allen Newell, Marvin Minsky
- 基本思想: 智能可以由内在的认知过程, 也即人的理性思维过程来解释。
Core idea: Intelligence can be explained by internal cognitive processes, that is, the rational thinking processes of human beings.
- mind $\Rightarrow$ computer, cognition $\Rightarrow$ information processing/internalization
- All kinds of intelligence $\Rightarrow$ Physical **symbolic computing** system
- Pros: Considering both introspectionism and Behaviorism. Making it possible to implement mind in computer experiments.
- Cons: 对心智的机械化描述。A mechanical image of mind/cognition.

CogSci Research Methodology

- **Externalism 外在主义方法**
- 1. Behaviorism 行为主义
- J.B. Watson 古典行为主义 (classical behaviorism) B.F. Skinner 斯金纳新行为主义 (Neobehaviorism)。
- 新行为主义把心理活动等同于行为本身的一组操作，认为用科学的操作来规定心理学上术语的意义，可以减少无谓的争论。
To solve the problem of unobservable mental concepts (like "hunger" or "anxiety"), neobehaviorists argued that a term is only meaningful if it is defined by the exact *operations* used to measure it.
- E.g., “hunger” as an internal feeling $\Rightarrow$ "the state of an organism after 24 hours of food deprivation."
 - 斯金纳箱实验：操作性条件反射 Operant Conditioning，增加对行为的强化与惩罚。
- 排斥了整体心理的内在意义 (meaning)、目的 (purpose) 和动机 (motivation)。

CogSci Research Methodology

- Cognitive Context **认知语境**方法
- **Connectionism** 联结主义
- 代表人物：David Rumelhart, James McClelland, Geoffrey Hinton.
- 主要思想：强调简单的类神经元结构之间连接的重要性，认知是相互连接的神经元的相互作用。
Core idea: Emphasizing the importance of connections between simple neuron-like structures; **cognition is the interaction of interconnected neurons.**
- 认知过程就在于神经网络从初始状态到最后完成的稳定状态——动态过程。
 - 神经网络 neural networks, 并行分布式处理 parallel distributed processing.
 - 结构上的并行神经连接，算法上并行计算，功能上并行信息处理。
 - 强调认知的内在整体机制 Internal “wholeness” mechanism.

What is CogSci?

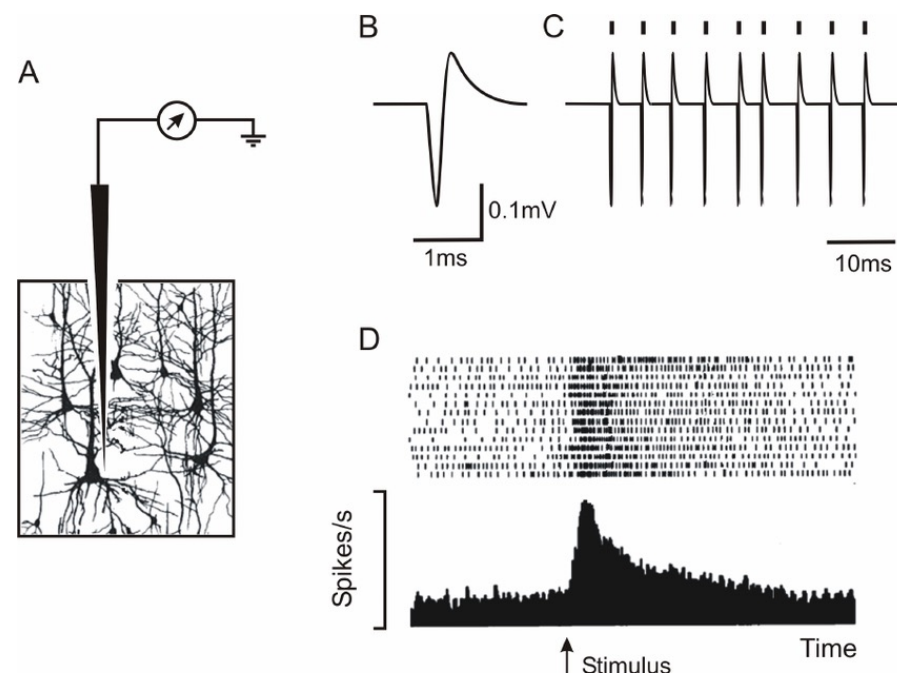
- Historical Background
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 - Tools; Experimental designs; Math and statistics.

Research Methods and Technology

- Brain-related Technology 大脑相关技术
 - 单细胞记录 Single-Unit Recording
 - 正电子发射层析摄影/断层扫描 Positron Emission Tomography (PET)
 - 磁共振成像 Magnetic Resonance Imaging (MRI)
 - 脑磁图 Magneto-Encephalography (MEG)
 - 脑电图 Electroencephalogram (EEG)
 - 脑机接口 Brain Computer Interface
- Behavioral Studies 行为研究
 - Stroop, N-back paradigms

Single-unit recording 单细胞记录

- **Single-unit recording** is a neuroscientific technique used to measure the electrical activity of individual neurons by inserting a microelectrode (微电极) into the brain to detect action potentials
- 现代脑机接口的前身



DOI: [10.4018/978-1-60960-021-1.ch006](https://doi.org/10.4018/978-1-60960-021-1.ch006)

Single-unit recording 单细胞记录

- **Limitations**
- Invasiveness: requires opening the skull to implant electrodes
- Limited scope: only records from a small number of neurons at a time
- Sampling bias: favor recording from larger neurons with larger electrical signals

Positron Emission Tomography (PET)

- Positron: 正电子; Emission: 发射层; Tomography: 层析摄影/断层扫描
- 原理:
 - 正电子是由某些放射性物质释放的微粒子。
Positrons are particles emitted by certain radioactive substances.
 - 将有放射性标记的液体**注射进体内**，迅速聚集在大脑的血管中。
A radiolabeled fluid is injected into the body, rapidly accumulating in the blood vessels of the brain.
 - 部分皮质兴奋时，带放射性标记的液体迅速移向兴奋区。
When specific cortical regions become active, the radiolabeled fluid swiftly shifts toward the activated zones.
 - 使用扫描装置测量放射性液体中的正电子数量。A scanning device is utilized to measure the quantity of positrons within the radioactive fluid.
 - 将扫描信息转换为代表大脑不同区域兴奋水平的图像。The scanned data is converted into images representing the activation levels across different brain regions.

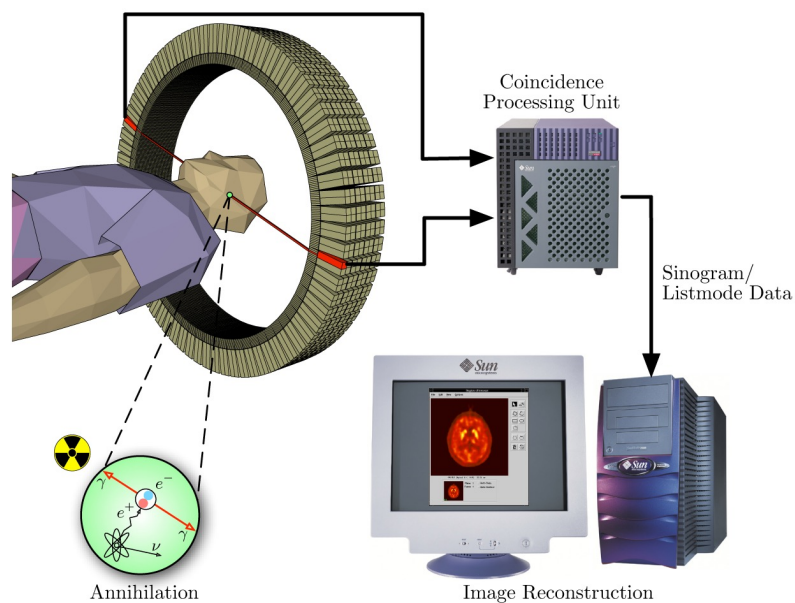
Source: https://en.wikipedia.org/wiki/Positron_emission_tomography

PET Examples

Isotopes used in PET scans

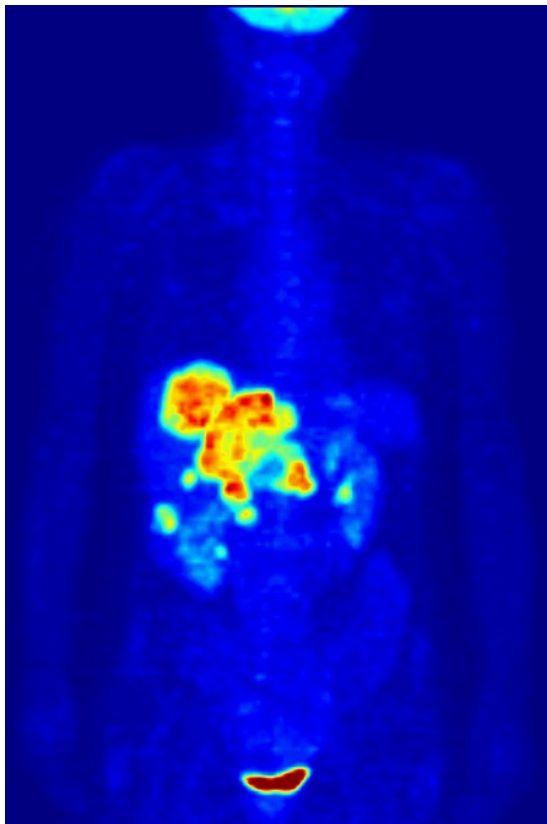
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Isotope	^{11}C	^{13}N	^{15}O	^{18}F	^{68}Ga	^{64}Cu	^{52}Mn	^{55}Co	^{89}Zr	^{82}Rb
Half-life	20 min	10 min	2 min	110 min	67.81 min	12.7 h	5.6 d	17.5 h	78.4 h ^[48]	1.3 min



Source: https://en.wikipedia.org/wiki/Positron_emission_tomography

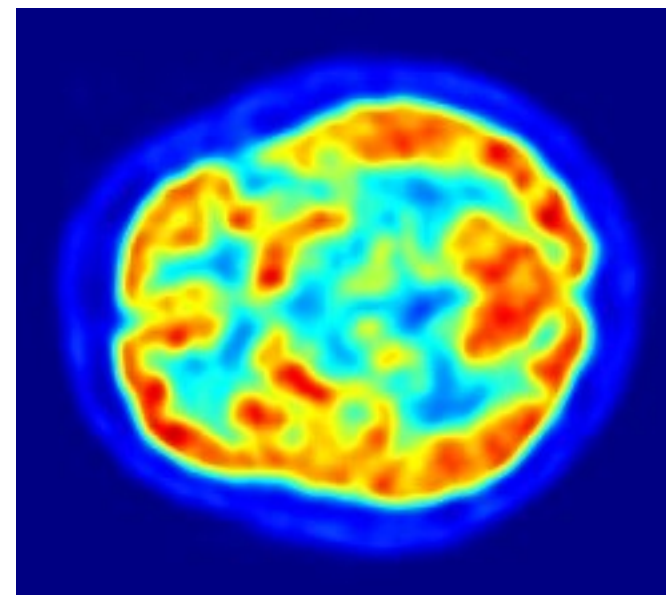
PET Examples



Whole-body PET scan using ^{18}F -FDG.

Fluorine-18 (^{18}F): Attached to glucose to form **FDG** (Fluorodeoxyglucose).

氟-18 (^{18}F): 与葡萄糖结合形成 FDG (氟代脱氧葡萄糖)。



A static PET scan of the human brain.

Positron Emission Tomography (PET)

- **Pro:** Pretty good spatial resolution (3~4 mm)
- **Cons:**
 - **Low temporal resolution:** can measure brain activity within the duration of about 60s; way longer than most cognitive processes.
 - **Indirect** measurement of neural activities.
 - **Invasive** technique: subject needs to be injected with a radioactive isotope-labeled solution (放射性同位素标注液).

PET for CogSci Research

DOI: 10.1017/s0033291797004856.

- Baker et al., 1997, The interaction between mood and cognitive function studied with PET 使用PET研究情绪与认知功能之间的相互作用
- 方法：正常受试者在诱发的兴高采烈 (elation) 和抑郁情绪 (depression) 状态下执行语言流畅性任务。使用正电子发射断层扫描 (PET) 测量局部脑血流量 (rCBF) ，将其作为神经活动的指标。
subjects performed a verbal fluency task under induced states of elation and depression. Positron Emission Tomography (PET) was used to measure regional cerebral blood flow (rCBF) as an index of neural activity.
- 结果 Findings:
 - 在兴奋和抑郁两种情绪状态下，外侧眶额皮质的局部脑血流量 (rCBF) 均有所增加；
 - 在抑郁状态下，喙侧内侧前额叶皮质 (prefrontal cortex) 的局部脑血流量会减少。
 - 前扣带回 (Anterior cingulate cortex) 激活的减弱是抑郁情绪特有的表现。

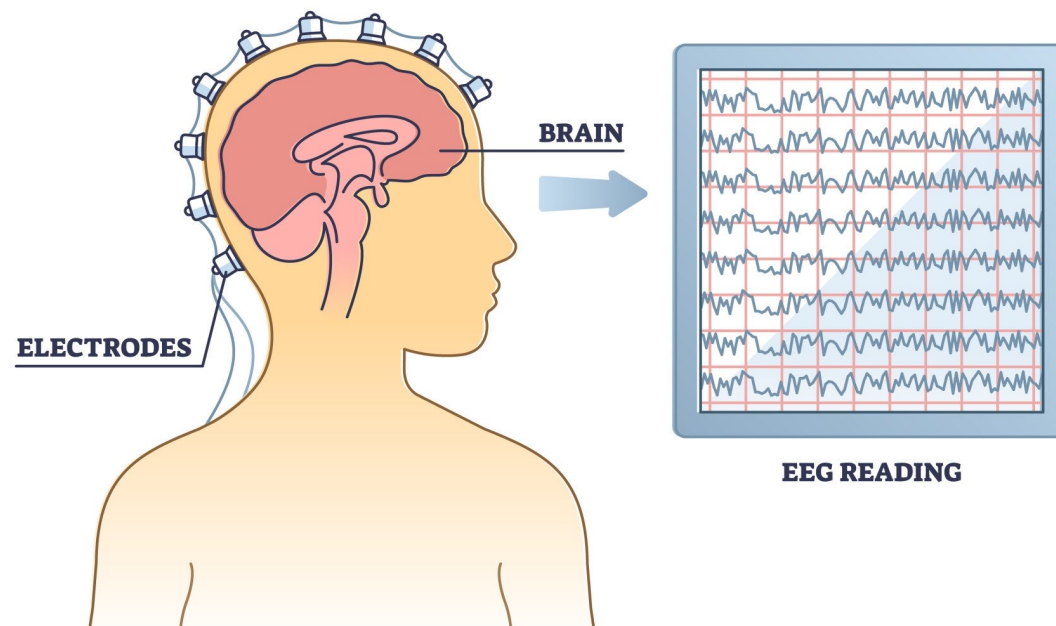
/ əˌlektrōənˈsefələˌɡrɑːm / *encephalo-* derived from Greek, meaning “brain”

Electroencephalogram, EEG 脑电图

- A technology that records the brain’s electrical activity

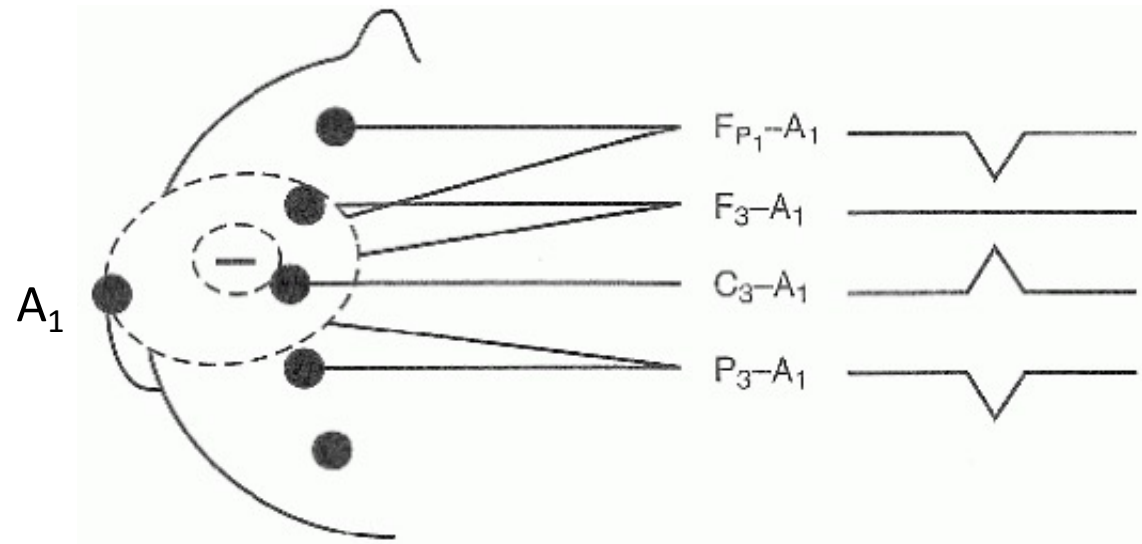
Placing electrodes (电极) on the scalp (头皮) to detect and measure the electrical signals produced by the brain.

Challenge: brain’s electrical activities are noisy, making it hard to track the relevant information processing procedures activated by certain stimuli



EEG Technology

- EEG does not directly measure the voltage
- EEG uses the principle of **differential** amplification, or recording voltage differences between different points using a pair of electrodes that compares one active exploring electrode site with another neighboring or distant reference electrode.



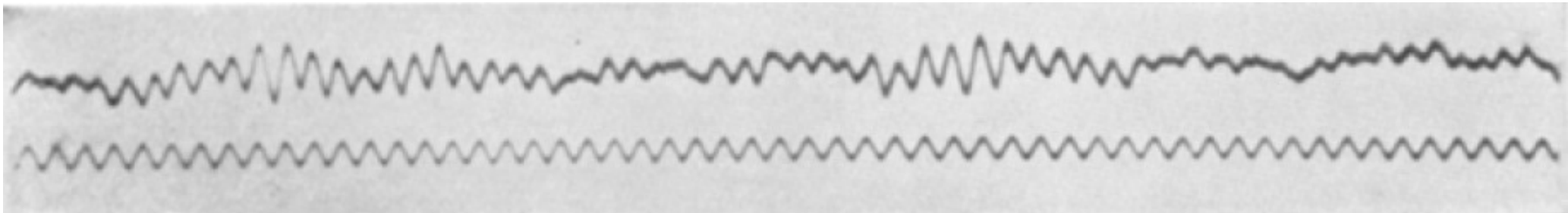
Source: <https://www.ncbi.nlm.nih.gov/books/NBK390346/>

EEG and ERP

- Event Related Potentials (ERPs)
- ERPs are extracted from EEG recordings by averaging the brainwave patterns associated with multiple presentations of the same event.
- This averaging process can **filter** out background noise and **isolate** the brain's response to the specific event.

EEG for CogSci Research

- First human EEG recording obtained by German physiologist Hans Berger in 1924



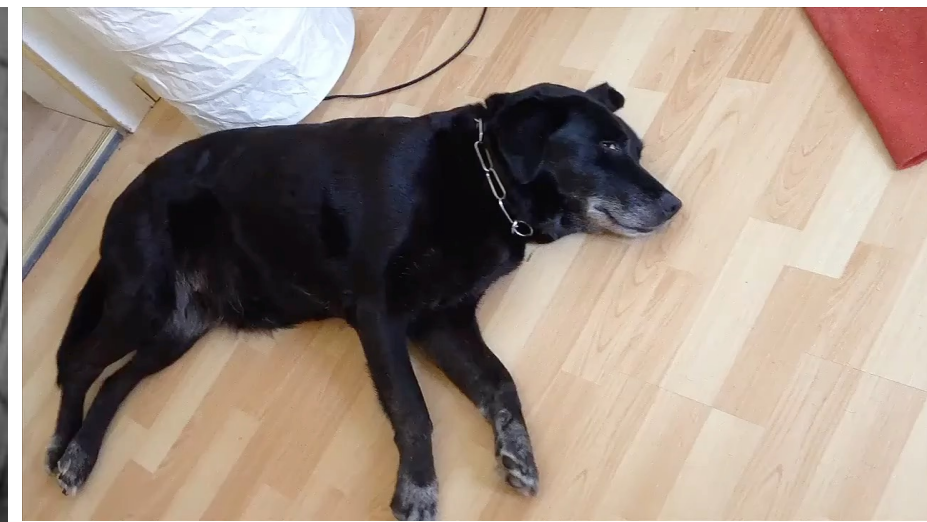
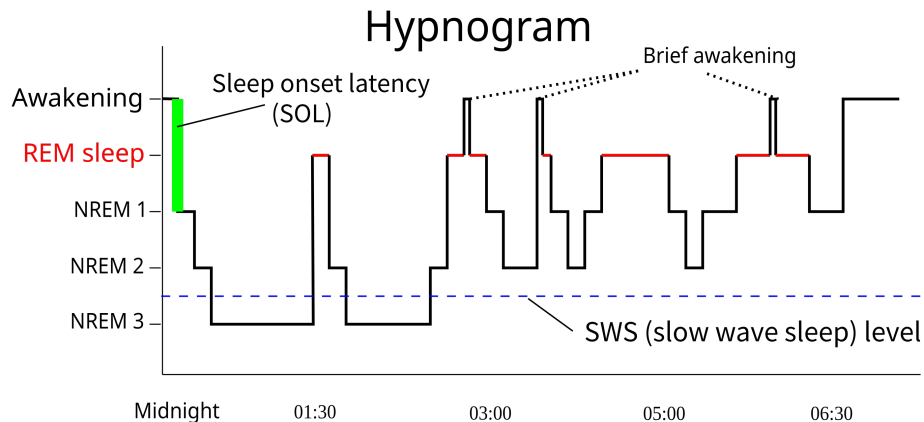
The upper tracing is EEG, and the lower is a 10 Hz timing signal.

Source: https://en.wikipedia.org/wiki/Electroencephalography_-_History

EEG for CogSci Research

Source: https://en.wikipedia.org/wiki/Rapid_eye_movement_sleep

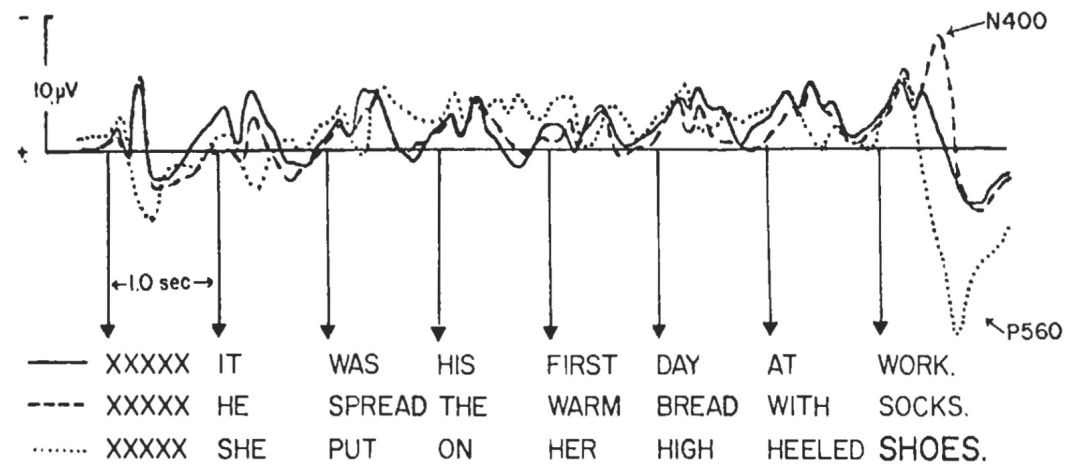
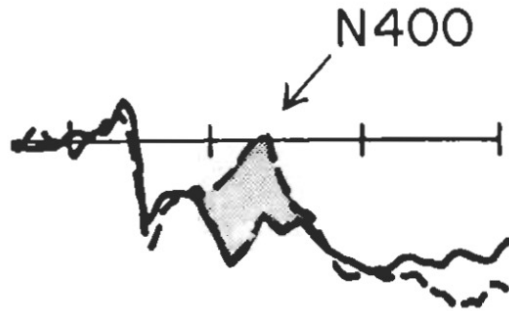
- EEG for monitoring stages of sleep
- Rapid eye movement sleep (REMS 快速动眼睡眠) is a unique phase of sleep in mammals (including humans) and birds.
 - characterized by random rapid movement of the eyes and the propensity to dream vividly.



EEG of sleep showing sleep cycles

EEG for CogSci Research

- Event related potentials during reading - **N400**
- Kutas and Hillyard, 1980: Unexpected ending of sentence elicits an N400 peaks in EEG signals.
- N400: a negative-going deflection that peaks around 400 milliseconds post-stimulus onset



DOI: <https://doi.org/10.1126/science.7350657>

EEG for CogSci Research

- EEG study of LLM usage
- “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”, MIT Media Lab
- “... LLM, Search Engine and Brain-only groups had significantly different neural connectivity patterns, reflecting divergent cognitive strategies ...”

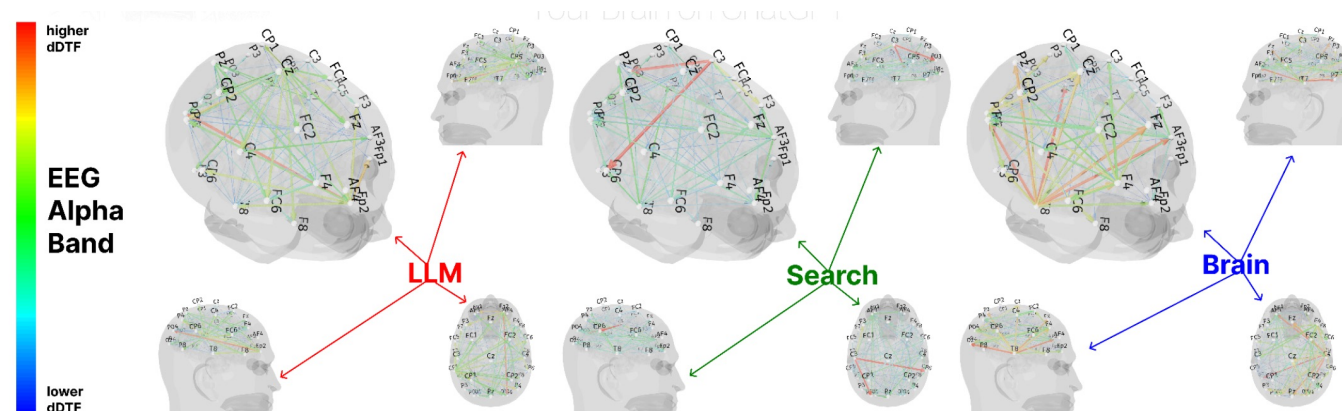
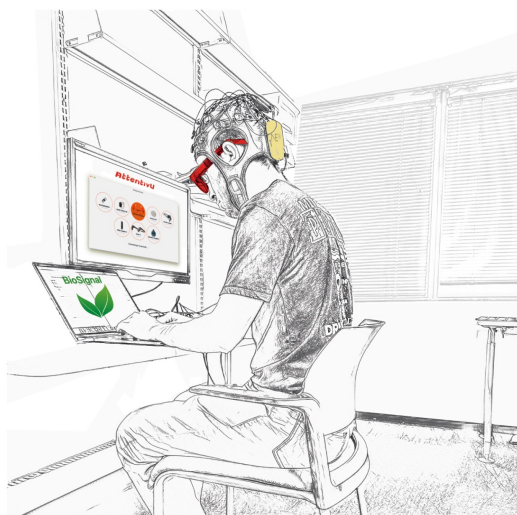


Figure 1. The dynamic Direct Transfer Function (dDTF) EEG analysis of Alpha Band for groups: LLM, Search Engine, Brain-only, including p-values to show significance from moderately significant (*) to highly significant (***) .

Magneto-Encephalography (MEG)

- MEG records the magnetic fields (磁场) produced by electrical currents occurring naturally in the brain, using sensitive magnetometers (磁场探测器).
- Arrays of SQUIDs (superconducting quantum interference devices 超导量子干扰装置) are currently the most common magnetometer
- MEG 对神经活动的测量精度高，因为脑颅对磁场不存在阻碍
- Resolution: better than EEG

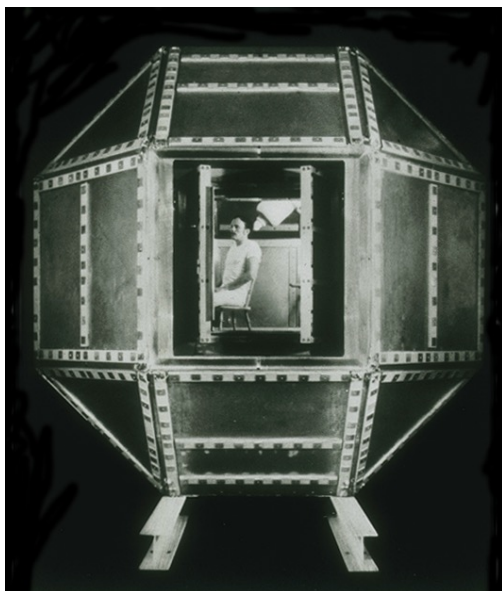


Source: <https://en.wikipedia.org/wiki/Magnetoencephalography>

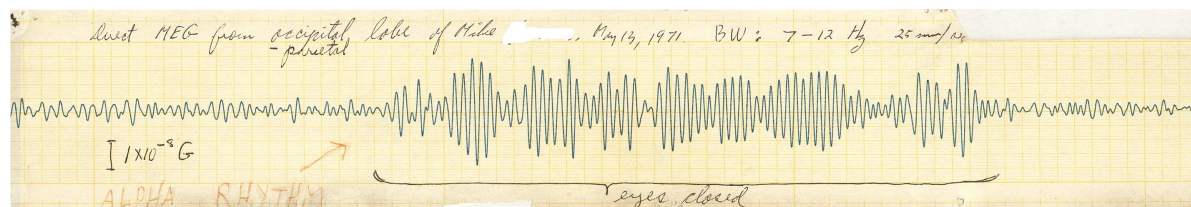
MEG Technology

Source: <https://en.wikipedia.org/wiki/Magnetoencephalography>

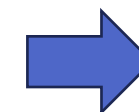
- First measured by David Cohen in 1968 (without SQUID); first SQUID-measured in 1970s at MIT.



Dr. Cohen's shielded room at MIT, in which first MEG was measured with a SQUID



First MEG measured with SQUID



Picture token in 2009

MEG Technology

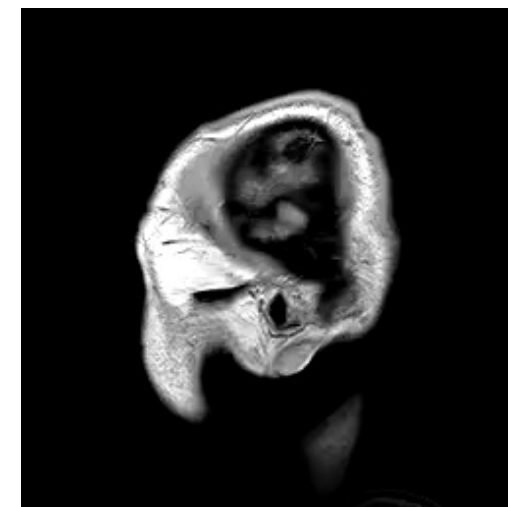
- Pros:
 - Directly measure brain activities. As a contrary, PET and fMRI only reflect the changes in blood flow.
 - Very good temporal resolution: millisecond level
- Cons:
 - The magnetic signals emitted by the brain are on the order of a few **femtoteslas** ($1 \text{ fT} = 10^{-15} \text{ T}$), shielding from external magnetic signals, including the Earth's magnetic field (地磁场 microtesla 10^{-5} T), is necessary.
 - Needs low temperature created by liquid nitrogen

Source: [https://en.wikipedia.org/wiki/Orders_of_magnitude_\(magnetic_field\)](https://en.wikipedia.org/wiki/Orders_of_magnitude_(magnetic_field))

Magnetic Resonance Imaging (MRI)

- **MRI**: An imaging technique that generates detailed, high-resolution images of the body's internal structures.
- Basic principle: use the interaction between **strong magnetic fields** (强磁场 1.5-3.0 T), **radiofrequency (RF) pulses** (射频脉冲, 300kHz-300GHz 电磁波), and the atoms in human body -- specifically, **hydrogen atoms** (protons, 氢原子/质子)

- 《核磁共振为何知道》 <https://www.bilibili.com/video/BV1di4y1y7au>
- Wiki: https://en.wikipedia.org/wiki/Magnetic_resonance_imaging



fMRI: Functional MRI

- A more sophisticated type of MRI that creates a **dynamic** record of metabolic activities over time, which is exclusively focused on the **brain**.
- 大脑局部神经元兴奋将引发该区域**血流量**增加，血液中含有**氧**和葡萄糖。
Localized neuronal activation in the brain triggers an increase in regional **blood flow**, which delivers **oxygen** and glucose.
- 血红蛋白所携氧的含量影响其磁场特性。
The amount of oxygen carried by hemoglobin affects its magnetic properties.
- fMRI能检测到大脑**功能性氧消耗**变化情况，清晰地显示高活动量区域的三维图像。
fMRI can detect changes in **functional oxygen consumption** within the brain, clearly displaying three-dimensional images of high-activity regions.

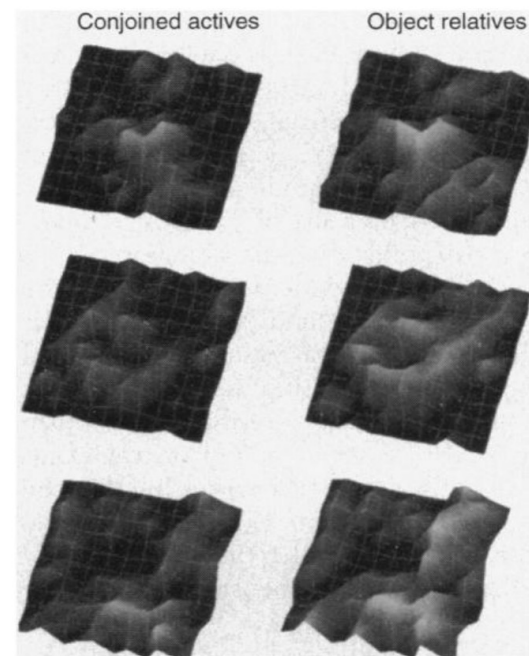
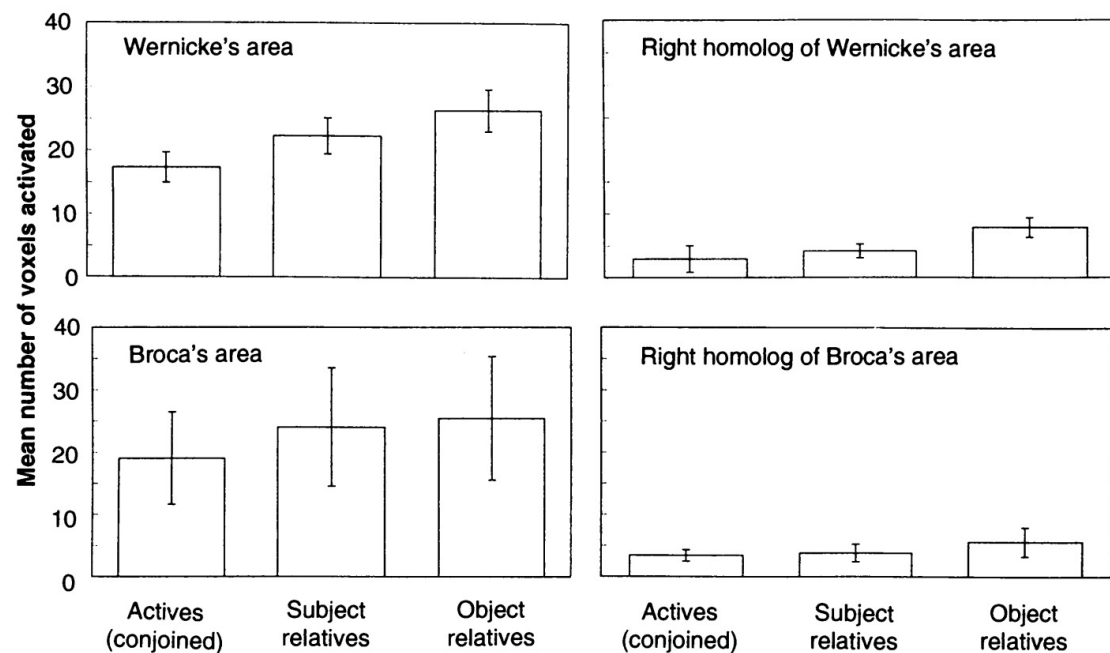
fMRI Technology

- Pros:
 - Safe: Non-invasive.
 - High spatial resolution: 1-2 mm.
- Cons:
 - Indirect measurement of neural activities.
 - Low temporal resolution: a few seconds, insufficient to trace cognitive activities.

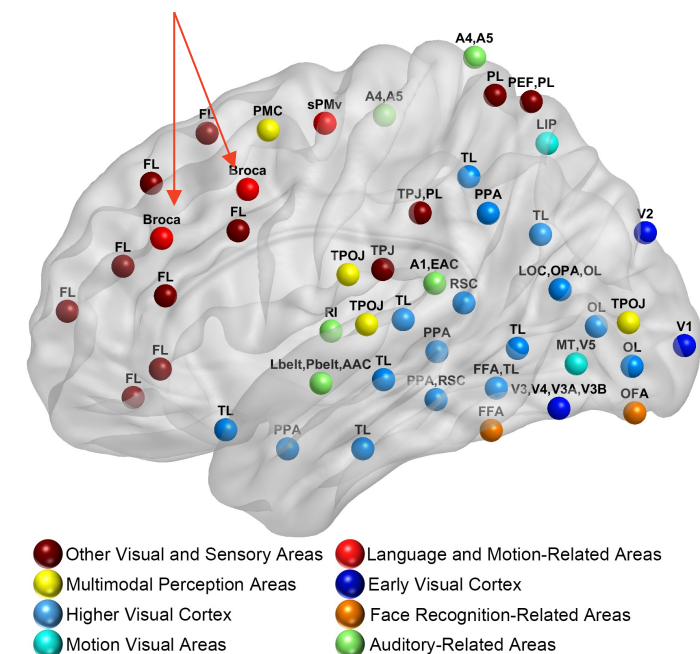
fMRI for CogSci Research

DOI: [10.1126/science.274.5284.114](https://doi.org/10.1126/science.274.5284.114)

- Just et al. (1996): Brain activation increases with the linguistic complexity of the sentences.



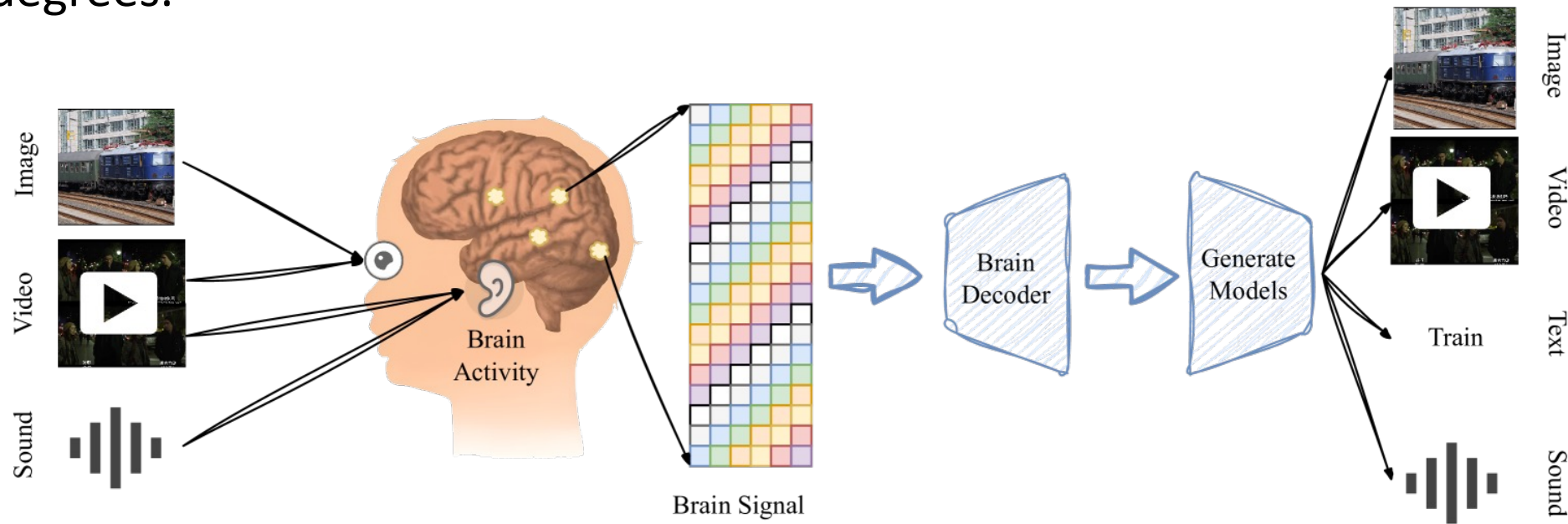
Broca's area



fMRI and AI Research

Source: <https://arxiv.org/html/2503.15978v1>

- Current AI approaches can decode brain signal measured by fMRI to some degrees.



After the human body receives external stimuli, the brain generates signals in specific regions. By decoding these brain signals and **using generative models for reconstruction**, the stimulus signals received by the brain are restored.

Common Experimental Designs

- **Stroop paradigm:** Proposed by John R. Stroop in 1935, to study the interference between two conflicting dimensions of a stimulus.
- Example: Participants need to name the ink color of a color word instead of reading the word itself.

Task 1: Read the words, ignoring the colors:

BLUE	ORANGE	YELLOW	RED	PURPLE
PINK	BLUE	BLACK	PURPLE	GREEN

Task 2: State the colors of the words, ignoring the actual text:

YELLOW	RED	PINK	BLUE	GREEN
PURPLE	YELLOW	BLUE	BLACK	PINK

Common Experimental Designs

- Task 2 **takes longer** than Task 1 (in most cases)
- Because human processing of word content (reading the text) is fast and automatic
- While the processing of color requires more deliberate control
- When the two conflict, the automatic processing (Task 1) will interfere the more complex task (Task 2), causing delay in response time and higher error rates.

Common Experimental Designs

- **N-back** paradigm: Tasks to study **working memory** by requiring participants to indicate if the current stimulus matches the one presented N trials earlier in a sequence. First introduced by Wayne Kirchner in 1958.
- Example: Experimenter read the following list of letters to the test subject:

T L H C H O **C** Q L **C** K **L** H C Q T R R K C H R

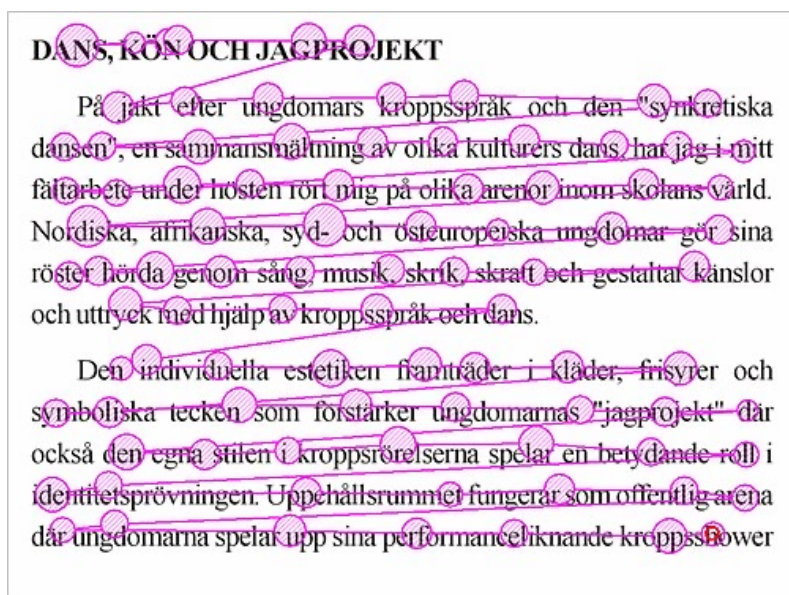
- The subject is supposed to indicate the letters in red circles as they were read 3 steps earlier.
- Can be used to measure the working memory capacity.

Source: <https://en.wikipedia.org/wiki/N-back>

Eye-tracking 眼动追踪

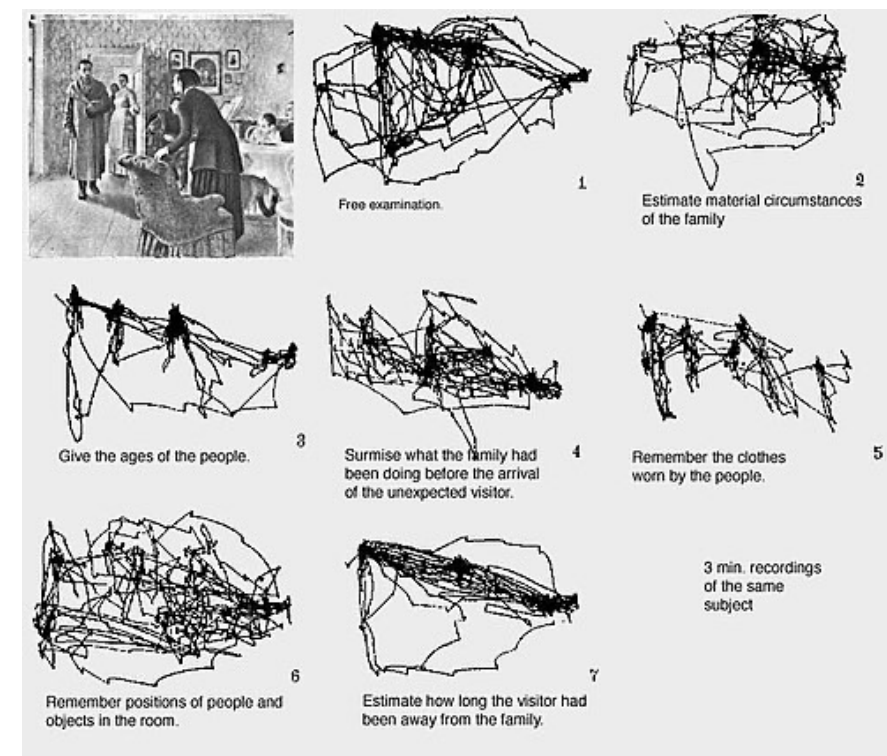
Source: https://en.wikipedia.org/wiki/Eye_tracking_-_Eye-tracking_vs._gaze-tracking

- Eye tracking: the process of measuring either the point of **gaze** (where one is looking) or the **motion** of an eye relative to the head. An eye tracker is a device for measuring eye positions and eye movement.



An example of fixations (注视) and saccades (扫视) over text.

Task given to a person influences his/her eye movement ⇒

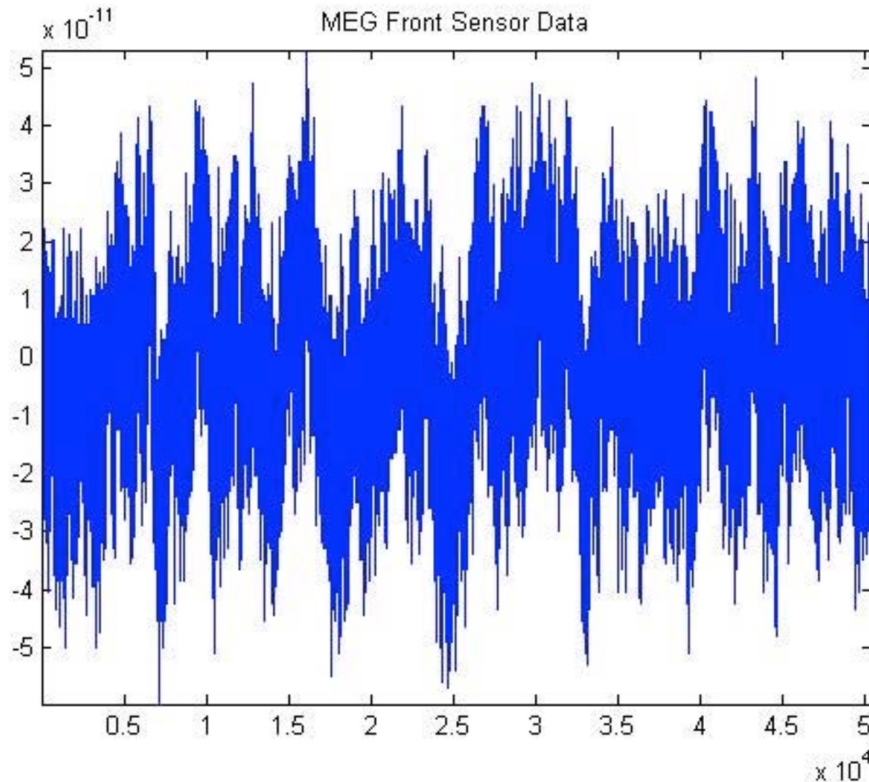


Math and Statistics for CogSci

- Data: Visualization and statistics
- Model
- Hypothesis Testing

Data Example

- On each channel of a magnetoencephalogram (脑磁图), we measure the magnetic field in an MEG scanner in the absence of a subject in order to characterize the background noise.



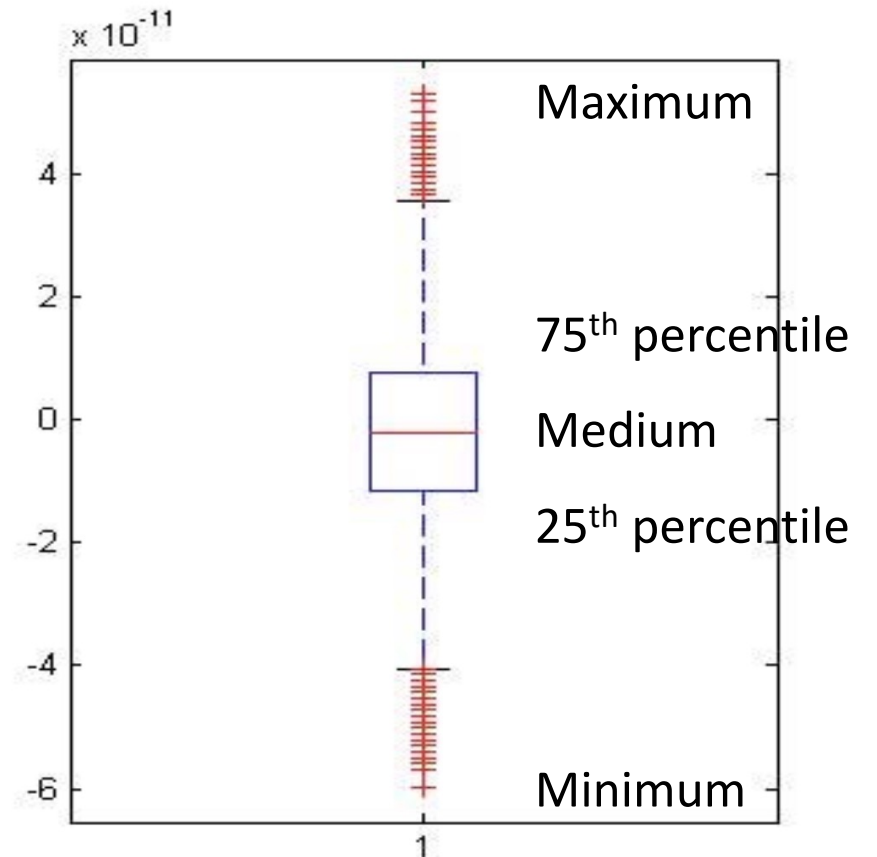
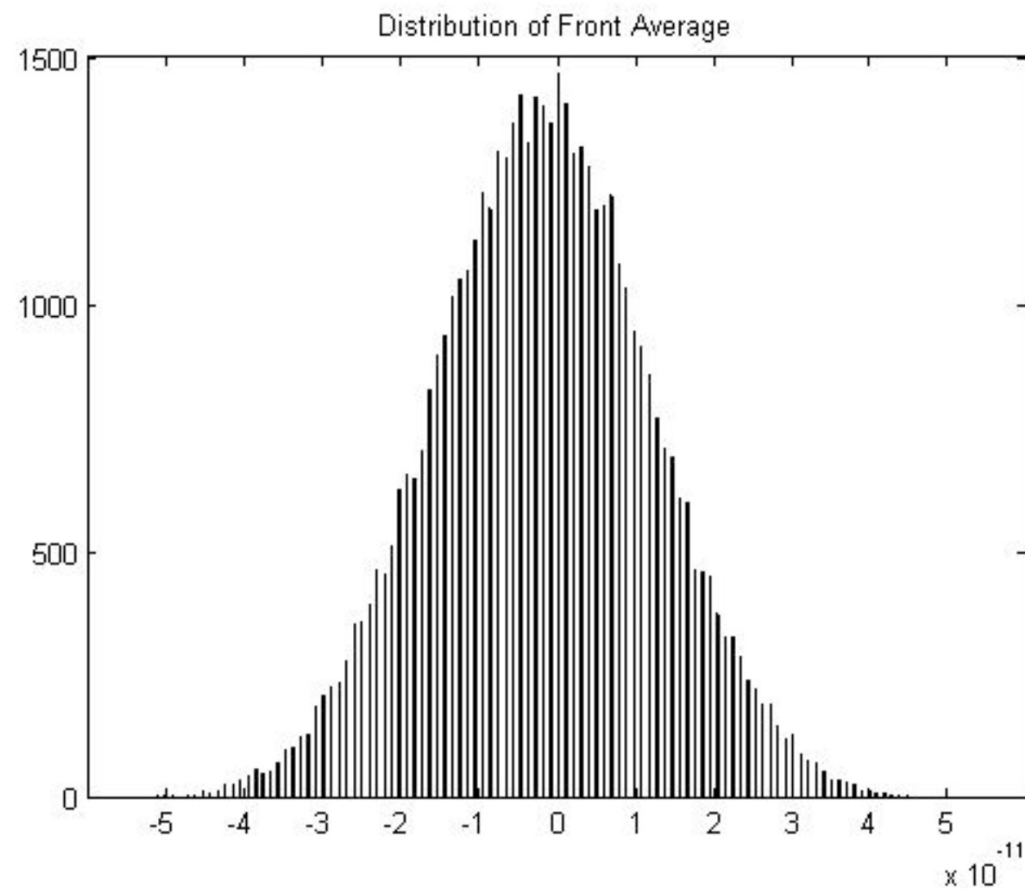
MEG data recorded during 14 min
X-axis: time $14 \times 60^2 \approx 5 \times 10^4$ sec.

Y-axis: order of 10^{-11} Tesla peak-to-peak.

Source: <https://ocw.mit.edu/courses/9-07-statistics-for-brain-and-cognitive-science-fall-2016/>, Lecture 3.

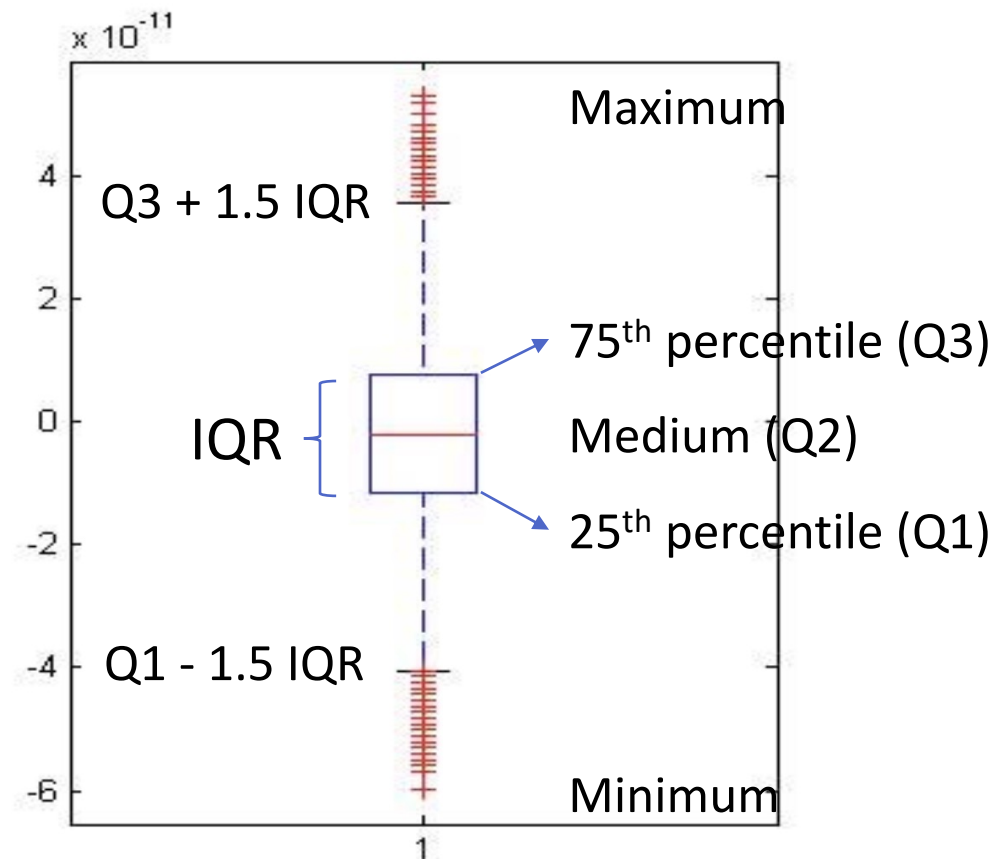
MEG Data Visualized

- Histogram and boxplot.



Five-Number Summary

- Five numbers:
 - Medium = 50th percentile (Q2)
 - 25th and 75th percentiles (Q1 and Q3)
 - Minimum and maximum (Q0 and Q4)
- Interquartile range (IQR):
 - Distance between 25th and 75th percentiles
- Whiskers:
 - Extended to 1.5 or 3 times IQR

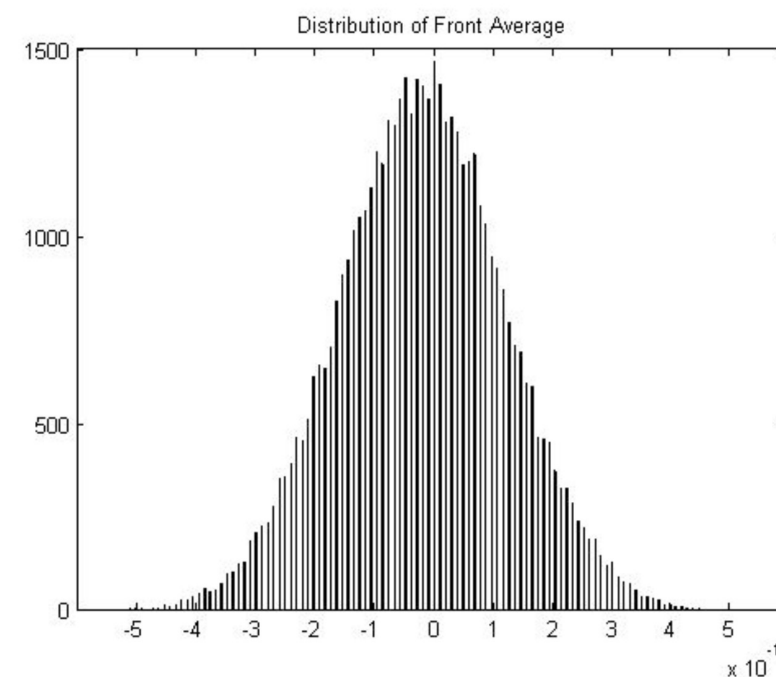


Gaussian Probability Model

- The shape of histogram indicates that a Gaussian model can be used to describe the distribution of data:

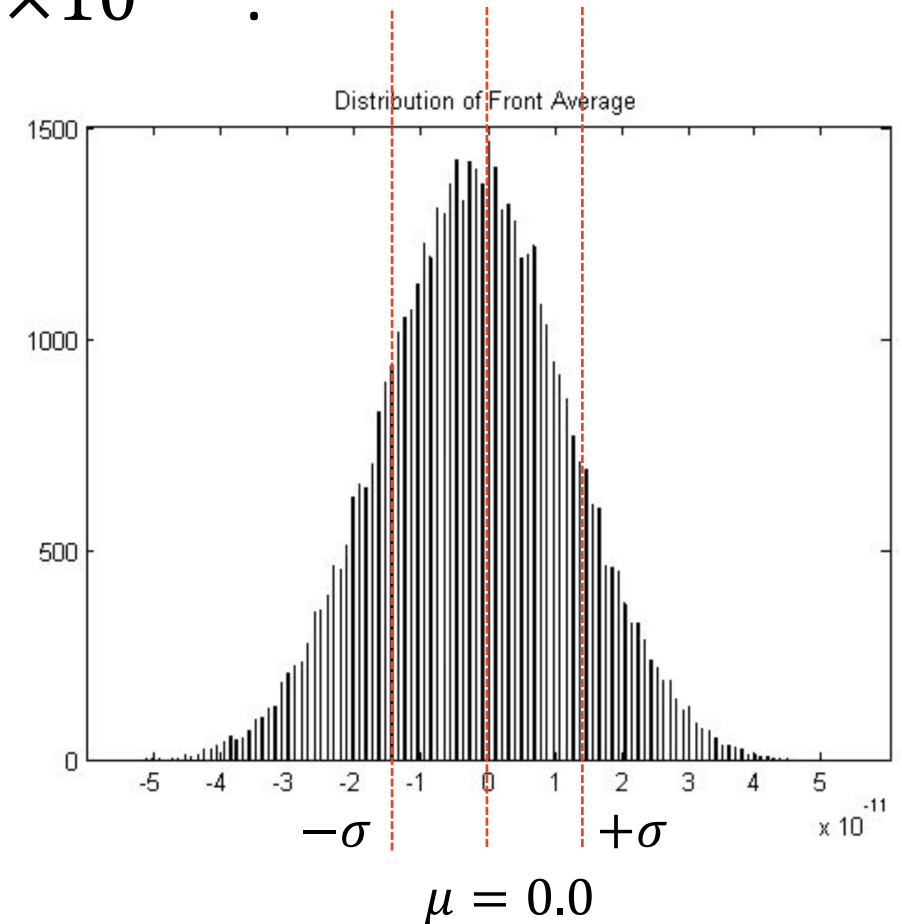
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- $f(x)$: probability density function;
- mean μ ; variance σ^2 .



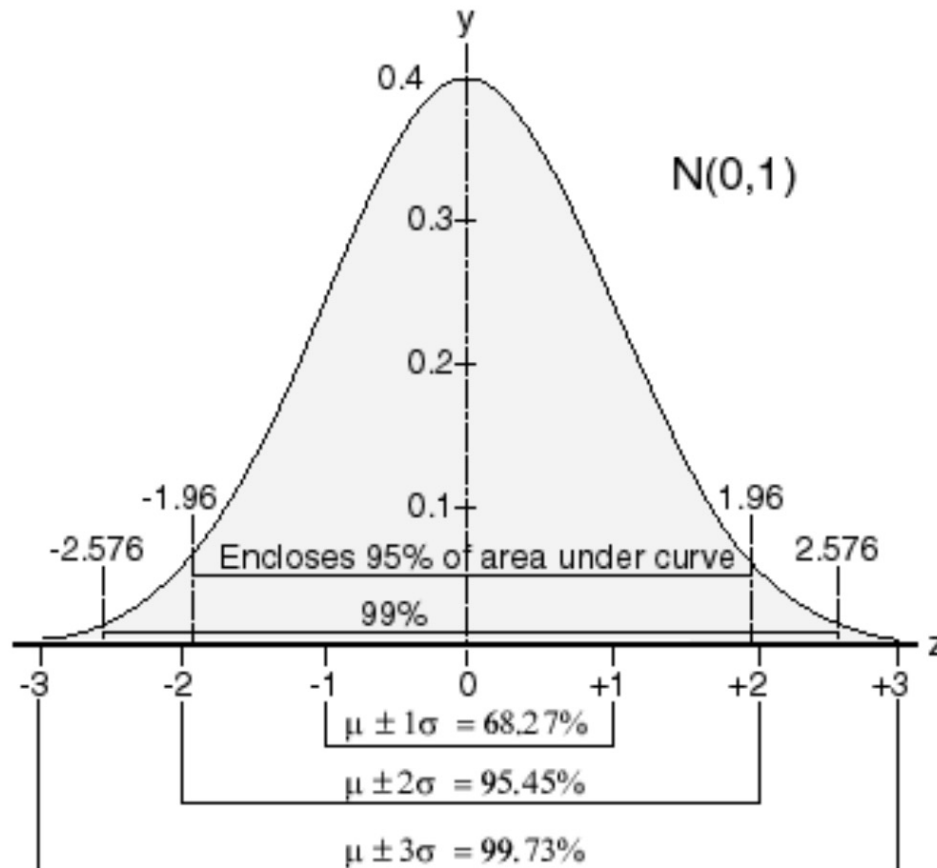
Gaussian model of MEG data

- Assume in the MEG data, $\mu = 0.0$, $\sigma = 1.4 \times 10^{-11}$.
- Interpreting σ : the average deviation from the mean that we expect to see.
- Not surprised to see a random variable X :
 $\mu - \sigma \leq X \leq \mu + \sigma$.
- Somewhat unusual to see:
 $X \leq \mu - 2\sigma$ or $X \geq \mu + 2\sigma$.



Approximated Rules from Gaussian

- The “2/3 to 0.95 Rule”.



$$P(\mu - \sigma \leq X \leq \mu + \sigma) = 0.67 \approx \frac{2}{3}$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$$

Hypothesis Testing

- **Hypothesis Testing 假设检验**
- Hypothesis: A statement (about a scientific question) in terms of a proposed value of a parameter in a probability model.
- Hypothesis testing is a process of establishing proof of *falsification*.
- Two components:
- **Null hypothesis H_0** : A stated value of parameter which defines the hypothesis that we want to reject.
- **Alternative hypothesis H_A** : A value (a set of values) different from that in H_0 , defining the hypothesis we wish to establish.

Analysis of MEG Sensor Bias

- Question: Could it be a bias in the MEG data?
- Assume we have a sample of data $x_1 \dots x_n$, and each data point is drawn from a Gaussian distribution $x_i \in \mathcal{N}(\mu, \sigma^2)$. We also assume that μ is unknown, and σ is known.
- If there is a bias, then $\mu \neq 0$:
- $\mu > 0$: positive bias; $\mu < 0$: negative bias.
- **Null hypothesis $H_0: \mu = 0$**
- **Alternative hypothesis $H_A: \mu > 0$ or $H_A: \mu < 0$**

Test the Hypothesis with a Statistic (统计量)

- A **test statistic**'s value will allow us to distinguish between the null and alternative hypothesis.
- In the MEG case, we choose sample mean $\bar{x} = \frac{x_1 + \dots + x_n}{n}$ as the statistic
- Recall that $\bar{x} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ (Central Limit Theorem):
- **Case I:** If $\bar{x} \gg 0$ or $\bar{x} \ll 0$, then we are willing to conclude $\mu \neq 0$, i.e., reject the null hypothesis H_0 .
- **Case II:** If $\bar{x}$ is close to 0, then we fail to reject H_0 .
(we do not say we accept H_0 because we do not reach a conclusion.)

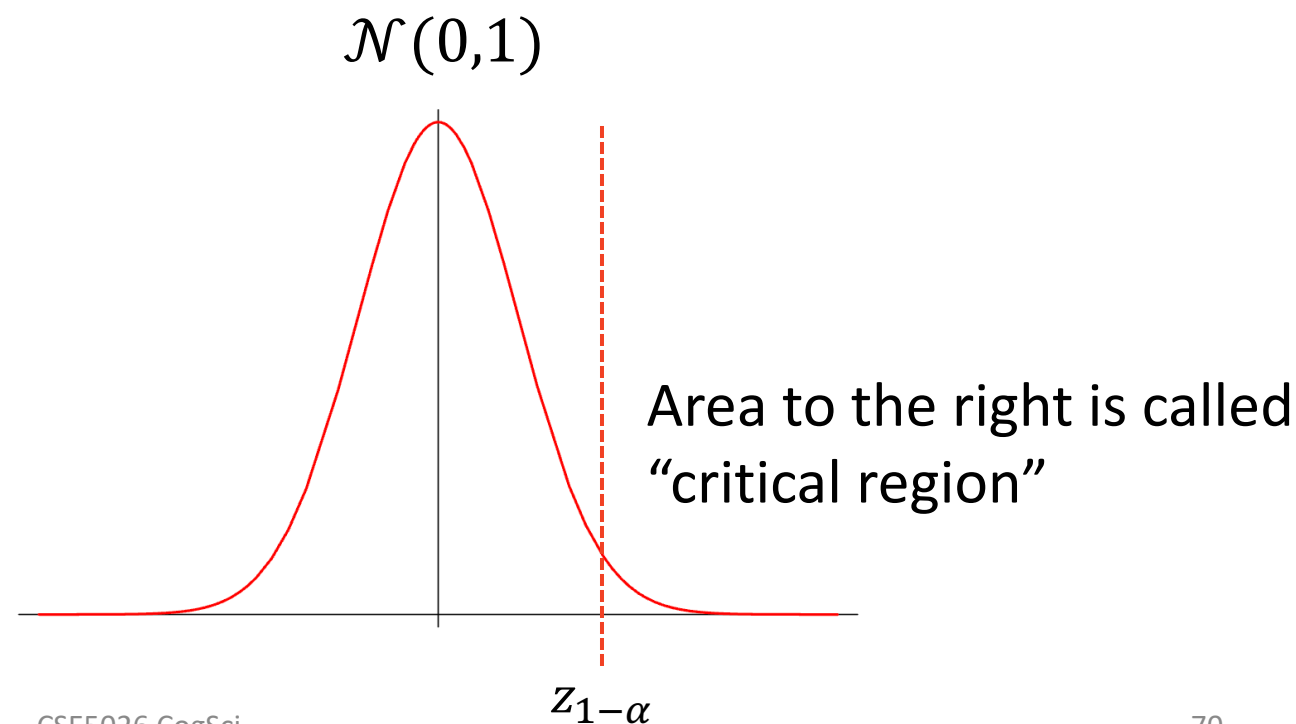
How correct are we to reject or not?

- In Case I, if we reject H_0 when it is true, then this is an error called **Type I error**.
- Let α be the probability of making this type of error:
- $\alpha = P(\text{rejecting } H_0 \mid H_0 \text{ is true}) \Rightarrow \alpha = P(\bar{x} > c_\alpha \mid \mu = 0)$
 - α describes the likelihood of observing an $\bar{x}$ that is greater than some cut-off value c_α , under the condition that $\mu = 0$
- Because $\bar{x} \sim \mathcal{N}(\mu, \sigma^2/n)$, α can be derived:
- $\bar{x}$ normalized to $z = \frac{\bar{x} - \mu}{\sqrt{\sigma^2/n}}$, and we have $z \sim \mathcal{N}(0,1)$, (go to next slide)

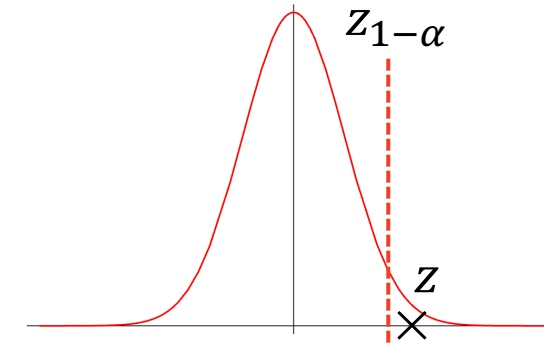
Type I Error

- (from previous slide) Then $\alpha = P(\bar{x} > c_\alpha \mid \mu = 0)$ is equivalent to:
- $\alpha = P(z > z_{1-\alpha})$, where $z_{1-\alpha}$ is the quantile of the standard Gaussian (normal) distribution.

α	$z_{1-\alpha}$
0.05	1.645
0.025	1.96
0.01	2.326



Example



- MEG bias data: Assume we have 500 observations $x_1 \dots x_n$, and the standard deviation is $\sigma = 1.1 \times 10^{-11}$ Tesla, and the sample mean is $\bar{x} = 0.11 \times 10^{-11}$ Tesla. Suppose we wish to test $H_0: \mu = 0$ against the alternative hypothesis $H_A: \mu > 0$ with $\alpha = 0.05$.
- Let's compute the z statistic: $z = \frac{\sqrt{n}(\bar{x}-0)}{\sigma} = \frac{\sqrt{500} \cdot (0.11)}{1.1} \approx 2.25$
- From the quantile table, we have $z_{1-\alpha} = z_{0.95} = 1.645$
- Then $z > z_{1-\alpha}$, meaning that we can reject and conclude that there is a positive bias in the MEG sensor.

Example (cont.)

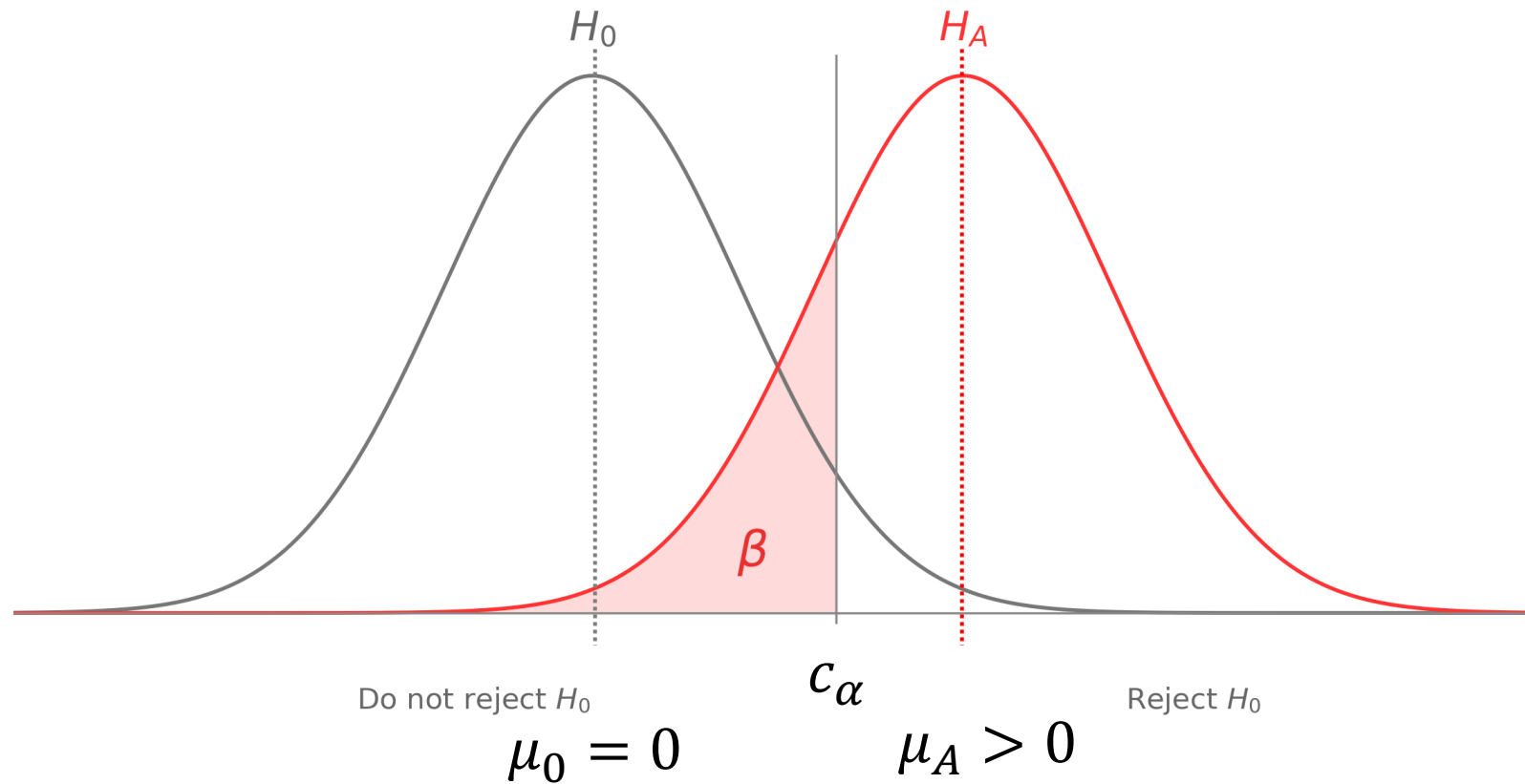
- We should also compute the p -value of z :
- $p = P(Z \geq z \mid H_0 \text{ is true}) = 1 - \Phi(z) = 1 - \Phi(2.25) \approx 0.012$,
- which means the probability of obtaining a Z statistics that is equal to or greater than $z = \frac{\sqrt{n}(\bar{x}-0)}{\sigma} = 2.25$.
- $p < \alpha = 0.05$ suggests that we can reject H_0 with fair confidence.

On the other hand

- **Case II:** If $\bar{x}$ is close to 0, then we fail to reject H_0 .
- If H_0 is not true, i.e., H_A is true and we fail to reject H_0 , then this is also an error -- **Type II error**.
- Let β be the probability of this type of error:
- $\beta = P(\text{fail to reject } H_0 \mid H_A \text{ is true})$, H_A is true means that $\mu = \mu_A$,
- $\beta = P(\bar{x} < c_\alpha \mid \mu = \mu_A) = P\left(\frac{\bar{x} - \mu_A}{\sqrt{\sigma^2/n}} < \frac{c_\alpha - \mu_A}{\sqrt{\sigma^2/n}}\right) = P(Z < z_\beta)$
- In practice, we care about $1 - \beta$, which is the **power** of the test (go to next slide)

Type II Error

- $\beta = P(\text{fail to reject } H_0 \mid H_A \text{ is true})$



Power of Test

- Power = $1 - \beta = P(\text{reject } H_0 \mid H_A \text{ is true})$
- Power = $P(\bar{x} > c_\alpha \mid \mu = \mu_A) = P(z > z_\beta)$
- We should not carry out a test if the power is not at least > 0.8 for important H_A .
- Should never report a negative result (failing to reject H_0) without reporting the power.

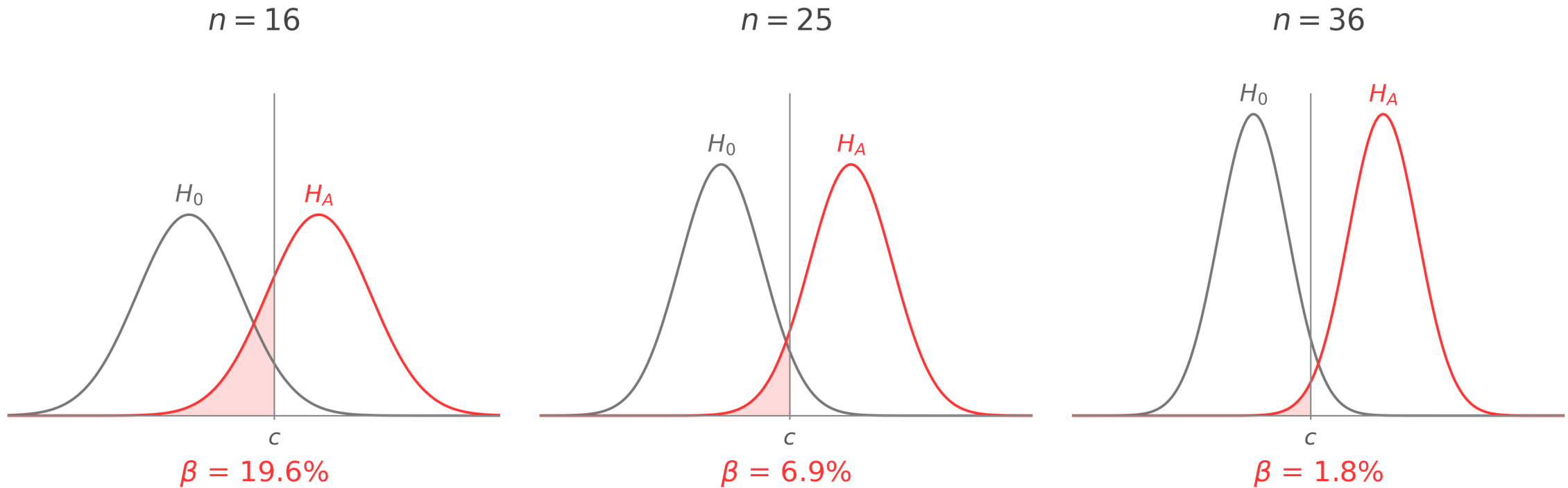
Power and sample size calculation

- Given $H_0: \mu = \mu_0, H_A: \mu = \mu_A > \mu_0$
- We will have power = $\Phi\left(\frac{\sqrt{n}|\mu_0 - \mu_A|}{\sigma} + z_\alpha\right)$, where Φ is the cumulative distribution function (CDF) of $\mathcal{N}(0,1)$.
- Then, power = $1 - \beta = \Phi\left(\frac{\sqrt{n}|\mu_0 - \mu_A|}{\sigma} + z_\alpha\right)$, [Refer to MIT9_07F16_lec12.pdf for detailed derivation.](#)
- $z_{1-\beta} = \frac{\sqrt{n}|\mu_0 - \mu_A|}{\sigma} + z_\alpha \Rightarrow z_{1-\beta} - z_\alpha = \frac{\sqrt{n}|\mu_0 - \mu_A|}{\sigma} = z_{1-\beta} + z_{1-\alpha}$
- Finally, $n = \frac{\sigma^2(z_{1-\beta} + z_{1-\alpha})^2}{\Delta^2}$, where $\Delta = |\mu_0 - \mu_A|$.

Power and sample size calculation

- Example: How large should $n = ?$ be so that we are at least 80% sure about the $H_A: \mu_A = 0.1 \times 10^{-11}$ Tesla, with $\alpha = 0.05$?
- Answer: Apply $z_{0.95} = 1.645$, $z_{0.80} = 0.84$, $\sigma = 1.1 \times 10^{-11}$ to the a
$$n = \frac{\sigma^2 (z_{1-\beta} + z_{1-\alpha})^2}{\Delta^2},$$
- $n = \frac{1.1^2 (1.645 + 0.84)^2}{0.1^2} \approx 748.$
- Round up to $n = 750$ instead of the original 500.

How sample size affects power



Future of Cognitive Science

- Artificial Intelligence 人工智能
 - LLMs, Agents, Robots
- Neuromorphic (Brain-inspired) Computing 类脑计算
- Brain Computer Interface 脑机接口